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End Semester Examination 2025

Name of the Course: B.Tech. (ECE)
 Name of the Paper : Computer Architecture
 Time: 3 Hours

Semester: VIII
 Paper Code : TOE810
 Maximum Marks: 100

Note:-

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question
- Total marks in each main question are twenty.
- Each question carries 10 marks

Q1	(20marks)	
(a)	Explain the function and capacity of following registers in a basic computer – Program Counter, Stack Pointer, instruction register, address register, accumulator.	CO1
(b)	Differentiate between direct, indirect and immediate addressing modes with the help of an example for each.	
(c)	Draw block diagram of a digital computer and explain its various components.	
Q2	(20 marks)	
(a)	Differentiate between the following instructions with examples (i) LDA and STA (ii) LXI and LDAX	CO1 CO2
(b)	Explain selective clear, insert and mask operations with the help of an example for each.	
(c)	Explain 16 bit instruction code for memory reference, register reference and input output reference instructions.	
Q3	(20 marks)	
(a)	Explain the concept of subroutine with example and discuss the instructions associated with it.	CO3
(b)	Write detailed notes on RAM and ROM	
(c)	Draw & explain the flowchart for instruction cycle.	
Q4	(20 marks)	
(a)	Explain flow of interrupt cycle with the help of a suitable flow chart	CO3 CO5
(b)	Explain memory hierarchy with the help of block diagram. Write detailed notes on Cache memory.	
(c)	Write short notes on Direct Memory Access	
Q5	(20 marks)	
(a)	What are four important steps in an instruction cycle. Explain instruction pipeline with the help of timing diagram for a program having 9 instructions with 4 steps each in an instruction.	CO4 CO5 CO6
(b)	Design Binary Coded Decimal adder combinational circuit using truth table and corresponding Boolean function	
(c)	Draw hypercube interconnection structures for (i) $n = 1$ (ii) $n = 2$ (iii) $n = 3$ Where n denotes number of neighbor nodes	

